

Convergence Analysis of RMSProp (Non-Convex, Smooth)

Algorithm Name

RMSProp (without bias correction; per-coordinate exponential moving average of squared gradients; per-coordinate normalization by the square root of the running average).

Mathematical Formulation

We consider minimizing a differentiable (possibly non-convex) function $f : \mathbb{R}^d \rightarrow \mathbb{R}$. Let $\{w_t\}_{t \geq 0} \subset \mathbb{R}^d$ denote the iterates. RMSProp maintains a running average of squared gradients

$$v_t = \beta v_{t-1} + (1 - \beta) g_t \odot g_t, \quad t \geq 1,$$

where $g_t = \nabla f(w_t)$, $\beta \in [0, 1)$ is a fixed decay parameter, $v_0 \in \mathbb{R}_+^d$ is an initial positive vector, and $\odot$ denotes elementwise product. The update is

$$w_{t+1} = w_t - \alpha P_t g_t, \quad P_t = \text{diag}(v_t^{-1/2}),$$

where $v_t^{-1/2}$ is taken elementwise and $\alpha > 0$ is a base stepsize. In coordinates,

$$(v_t)_j = \beta(v_{t-1})_j + (1 - \beta)(g_t)_j^2, \quad (w_{t+1})_j = (w_t)_j - \alpha \frac{(g_t)_j}{\sqrt{(v_t)_j}}, \quad j = 1, \dots, d.$$

To avoid division-by-zero, assume $(v_0)_j > 0$ for all j ; the analysis below uses $(v_0)_j \geq 0$ and the fact that for $g_t \neq 0$, $(v_t)_j > 0$ automatically when $\beta < 1$.

Pseudocode

Input: $w_0 \in \mathbb{R}^d$, $v_0 \in \mathbb{R}_+^d$ (e.g., $v_0 = \delta \cdot \mathbf{1}$ with $\delta > 0$), base stepsize $\alpha > 0$, decay $\beta \in [0, 1)$, maximum iterations T , tolerance tol .

For $t = 1, 2, \dots, T$:

1. Compute gradient: $g_t \leftarrow \nabla f(w_t)$
2. Update squared-average: $v_t \leftarrow \beta v_{t-1} + (1 - \beta) g_t \odot g_t$
3. Compute preconditioner: $P_t \leftarrow \text{diag}(v_t^{-1/2})$
4. Update parameters: $w_{t+1} \leftarrow w_t - \alpha P_t g_t$
5. If $\|g_t\|_2 \leq \text{tol}$ then stop

Output: $w_{\text{out}} = w_{t^*}$ where t^* is the last iteration.

Dry Run Example

We minimize $f(w) = (w - 3)^2$ in $d = 1$ starting at $w_0 = 0$ with $\alpha = 0.1$, $\beta = 0.9$, $v_0 = 0$. The gradient is $g(w) = 2(w - 3)$.

Iteration 1:

$$g_1 = 2(0 - 3) = -6, \quad v_1 = 0.9 \cdot 0 + 0.1 \cdot 36 = 3.6, \quad w_1 = 0 - 0.1 \cdot \frac{-6}{\sqrt{3.6}} \approx 0.316227766.$$

Objective: $f(w_1) = (0.316227766 - 3)^2 = 9 - 2 \cdot 3 \cdot 0.316227766 + 0.1 \approx 7.202633404$.

Iteration 2:

$$g_2 = 2(0.316227766 - 3) \approx -5.367544468, \quad g_2^2 \approx 28.810533616,$$

$$v_2 = 0.9 \cdot 3.6 + 0.1 \cdot 28.810533616 \approx 6.121053362, \quad w_2 = 0.316227766 - 0.1 \cdot \frac{-5.367544468}{\sqrt{6.121053362}} \approx 0.533158166.$$

Objective: $f(w_2) \approx (0.533158166 - 3)^2 \approx 6.085309$.

Iteration 3:

$$g_3 = 2(0.533158166 - 3) \approx -4.933683668, \quad g_3^2 \approx 24.341234,$$

$$v_3 = 0.9 \cdot 6.121053362 + 0.1 \cdot 24.341234 \approx 7.9430715, \quad w_3 = 0.533158166 - 0.1 \cdot \frac{-4.933683668}{\sqrt{7.9430715}} \approx 0.708217.$$

Objective: $f(w_3) \approx (0.708217 - 3)^2 \approx 5.25228$.

Assumptions

We study the deterministic setting $g_t = \nabla f(w_t)$ under:

1. f is lower bounded: $\inf_{w \in \mathbb{R}^d} f(w) =: f_\star > -\infty$.
2. f is L -smooth: $\|\nabla f(x) - \nabla f(y)\|_2 \leq L \|x - y\|_2$ for all $x, y \in \mathbb{R}^d$ (equivalently, ∇f is L -Lipschitz).
3. Bounded gradients in ℓ_∞ : there exists $G_\infty > 0$ such that $\|\nabla f(w)\|_\infty \leq G_\infty$ for all iterates w produced by the algorithm.
4. RMSProp parameters: $\beta \in [0, 1)$ is fixed, $v_0 \in \mathbb{R}_+^d$ with $(v_0)_j \geq 0$ for all j (for well-definedness we may take $(v_0)_j > 0$).
5. Base stepsize $\alpha > 0$ (possibly chosen as a function of the horizon T).

Main Convergence Theorem

Let $V := \max\{G_\infty^2, \max_j (v_0)_j\}$. Under the assumptions above, the RMSProp iterates satisfy, for any $T \geq 1$,

$$\min_{1 \leq t \leq T} \|\nabla f(w_t)\|_2^2 \leq \frac{G_\infty}{\alpha T} (f(w_1) - f_\star) + \frac{G_\infty \alpha L d}{2(1 - \beta)}.$$

In particular, choosing $\alpha = \sqrt{\frac{2(1-\beta)(f(w_1)-f_\star)}{LdG_\infty T}}$ yields

$$\min_{1 \leq t \leq T} \|\nabla f(w_t)\|_2^2 \leq G_\infty \sqrt{\frac{2Ld(f(w_1)-f_\star)}{(1-\beta)T}} = \mathcal{O}\left(\frac{1}{\sqrt{T}}\right).$$

Theoretical Properties

- **Stability:** The per-iteration scaled step satisfies $\|P_t g_t\|_2^2 = \sum_{j=1}^d \frac{(g_t)_j^2}{(v_t)_j} \leq \frac{d}{1-\beta}$, uniformly in t .
- **Descent coefficient:** The effective curvature along g_t is at least $\lambda_{\min}(P_t) \geq 1/\sqrt{V} \geq 1/G_\infty$, ensuring a uniform descent component along the gradient direction.
- **Hyperparameter sensitivity:** Larger β increases memory but also the factor $1/(1-\beta)$ in the variance term, which degrades constants in the rate bound.
- **Comparison to SGD:** Compared to SGD with stepsize η , RMSProp adaptively rescales coordinates by $1/\sqrt{v_t}$, yielding bounded effective step energy $\|P_t g_t\|_2^2$, which can be advantageous under gradient anisotropy.

Discussion

The bound shows the standard nonconvex rate $\mathcal{O}(1/\sqrt{T})$ on the minimum squared gradient norm with explicit constants highlighting the roles of $(1-\beta)$, dimension d , smoothness L , and the ℓ_∞ gradient bound G_∞ . The proof is deterministic; expectations are therefore trivial if introduced (i.e., $\mathbb{E}[\cdot] = (\cdot)$).

Preliminaries (Definitions and Lemmas)

We collect standard smoothness and RMSProp-specific bounds needed in the proof.

Smoothness inequality. If f is L -smooth, then for any $x, s \in \mathbb{R}^d$,

$$f(x+s) \leq f(x) + \langle \nabla f(x), s \rangle + \frac{L}{2} \|s\|_2^2. [\text{Step1}] \quad (\text{P1})$$

Bounds on v_t . Let $g_t = \nabla f(w_t)$ and $v_t = \beta v_{t-1} + (1-\beta)g_t \odot g_t$ with $\beta \in [0, 1)$.

- **Upper bound:** For each coordinate j ,

$$0 \leq (v_t)_j \leq \max \left\{ (v_0)_j, \sup_{s \leq t} (g_s)_j^2 \right\} \leq \max \{ (v_0)_j, G_\infty^2 \} \leq V. [\text{Step2}] \quad (\text{P2})$$

Proof: By induction in t . If $(v_{t-1})_j \leq V$ and $(g_t)_j^2 \leq G_\infty^2 \leq V$, then $(v_t)_j = \beta(v_{t-1})_j + (1-\beta)(g_t)_j^2 \leq \beta V + (1-\beta)V = V$. Nonnegativity is obvious since both terms are nonnegative. [Step 2]

- Lower bound relative to g_t : For each j ,

$$(v_t)_j = \beta(v_{t-1})_j + (1-\beta)(g_t)_j^2 \geq (1-\beta)(g_t)_j^2, [Step3] \quad (P3)$$

hence for all j ,

$$\frac{(g_t)_j^2}{(v_t)_j} \leq \frac{1}{1-\beta}. [Step4] \quad (P4)$$

Matrix bounds for the preconditioner. Let $P_t = \text{diag}(v_t^{-1/2})$. Then P_t is positive definite as long as $(v_t)_j > 0$ for all j . Moreover, using (P2) we have

$$\lambda_{\min}(P_t) = \min_j \frac{1}{\sqrt{(v_t)_j}} \geq \frac{1}{\sqrt{V}} \geq \frac{1}{G_\infty}. [Step5] \quad (P5)$$

Consequently, for any $x \in \mathbb{R}^d$,

$$x^\top P_t x \geq \lambda_{\min}(P_t) \|x\|_2^2 \geq \frac{1}{\sqrt{V}} \|x\|_2^2 \geq \frac{1}{G_\infty} \|x\|_2^2. [Step6] \quad (P6)$$

Also, by (P4),

$$\|P_t g_t\|_2^2 = \sum_{j=1}^d \frac{(g_t)_j^2}{(v_t)_j} \leq \sum_{j=1}^d \frac{1}{1-\beta} = \frac{d}{1-\beta}. [Step7] \quad (P7)$$

Main Proof (Step-by-Step Derivation)

We prove the main descent inequality per iteration and then sum across iterations.

Let $s_t := -\alpha P_t g_t$. By L -smoothness (P1) with $x = w_t$ and $s = s_t$,

$$f(w_{t+1}) = f(w_t + s_t) \leq f(w_t) + \langle \nabla f(w_t), s_t \rangle + \frac{L}{2} \|s_t\|_2^2. [Step8] \quad (M1)$$

Since $s_t = -\alpha P_t g_t$ and $g_t = \nabla f(w_t)$ (deterministic setting),

$$\langle \nabla f(w_t), s_t \rangle = -\alpha \langle g_t, P_t g_t \rangle = -\alpha g_t^\top P_t g_t, [Step9] \quad (M2)$$

$$\|s_t\|_2^2 = \alpha^2 \|P_t g_t\|_2^2. [Step10] \quad (M3)$$

Substituting (M2)–(M3) into (M1) yields

$$f(w_{t+1}) \leq f(w_t) - \alpha g_t^\top P_t g_t + \frac{L\alpha^2}{2} \|P_t g_t\|_2^2. [Step11] \quad (M4)$$

By (P6) with $x = g_t$ (i.e., $g_t^\top P_t g_t \geq \frac{1}{\sqrt{V}} \|g_t\|_2^2 \geq \frac{1}{G_\infty} \|g_t\|_2^2$),

$$g_t^\top P_t g_t \geq \frac{1}{G_\infty} \|g_t\|_2^2, [Step12] \quad (M5)$$

and by (P7),

$$\|P_t g_t\|_2^2 \leq \frac{d}{1-\beta}. [Step13] \quad (M6)$$

Applying (M5)–(M6) to (M4) gives the per-iteration descent bound

$$f(w_{t+1}) \leq f(w_t) - \frac{\alpha}{G_\infty} \|g_t\|_2^2 + \frac{L\alpha^2}{2} \cdot \frac{d}{1-\beta}. [\text{Step14}] \quad (\text{M7})$$

Summing (M7) over $t = 1, \dots, T$ yields

$$f(w_{T+1}) - f(w_1) \leq -\frac{\alpha}{G_\infty} \sum_{t=1}^T \|g_t\|_2^2 + \frac{L\alpha^2}{2} \cdot \frac{d}{1-\beta} \cdot T. [\text{Step15}] \quad (\text{M8})$$

Since $f(w_{T+1}) \geq f_\star$,

$$f_\star - f(w_1) \leq -\frac{\alpha}{G_\infty} \sum_{t=1}^T \|g_t\|_2^2 + \frac{L\alpha^2}{2} \cdot \frac{d}{1-\beta} \cdot T, [\text{Step16}] \quad (\text{M9})$$

which rearranges to

$$\sum_{t=1}^T \|\nabla f(w_t)\|_2^2 \leq \frac{G_\infty}{\alpha} (f(w_1) - f_\star) + \frac{G_\infty L \alpha d}{2(1-\beta)} T. [\text{Step17}] \quad (\text{M10})$$

Dividing both sides by T and using $\min_{1 \leq t \leq T} a_t \leq \frac{1}{T} \sum_{t=1}^T a_t$ for nonnegative a_t ,

$$\min_{1 \leq t \leq T} \|\nabla f(w_t)\|_2^2 \leq \frac{1}{T} \sum_{t=1}^T \|\nabla f(w_t)\|_2^2 \leq \frac{G_\infty}{\alpha T} (f(w_1) - f_\star) + \frac{G_\infty L \alpha d}{2(1-\beta)}. [\text{Step18}] \quad (\text{M11})$$

This proves the first claimed bound in the Main Convergence Theorem. [Step 18]

Rate Derivation (With Constants)

Consider the upper bound in (M11) as a function of $\alpha > 0$:

$$\phi_T(\alpha) = \frac{A}{\alpha T} + B\alpha, \quad A := G_\infty(f(w_1) - f_\star), \quad B := \frac{G_\infty L d}{2(1-\beta)}.$$

Minimizing $\phi_T(\alpha)$ over $\alpha > 0$ gives

$$\alpha^\star = \sqrt{\frac{A}{BT}} = \sqrt{\frac{2(1-\beta)(f(w_1) - f_\star)}{L d G_\infty T}}, [\text{Step19}] \quad (\text{R1})$$

and the minimized value

$$\begin{aligned} \phi_T(\alpha^\star) &= \frac{A}{\alpha^\star T} + B\alpha^\star = 2\sqrt{\frac{AB}{T}} = 2\sqrt{\frac{G_\infty(f(w_1) - f_\star)}{T} \cdot \frac{G_\infty L d}{2(1-\beta)}} \\ &= G_\infty \sqrt{\frac{2 L d (f(w_1) - f_\star)}{(1-\beta) T}}. [\text{Step20}] \end{aligned} \quad (\text{R2})$$

Substituting $\alpha = \alpha^\star$ into (M11) yields

$$\min_{1 \leq t \leq T} \|\nabla f(w_t)\|_2^2 \leq \phi_T(\alpha^\star) = G_\infty \sqrt{\frac{2 L d (f(w_1) - f_\star)}{(1-\beta) T}}.$$

Special Cases

- $\beta = 0$ (no memory): Then $v_t = g_t \odot g_t$ and $P_t = \text{diag}(|g_t|^{-1})$, so the update reduces to coordinatewise sign descent $w_{t+1} = w_t - \alpha \text{sign}(g_t)$. The bound specializes with $(1 - \beta) = 1$.
- $d = 1$ (scalar case): The constants simplify to $B = \frac{G_\infty L}{2(1-\beta)}$ and the same $\mathcal{O}(1/\sqrt{T})$ rate holds.
- Large $\beta \uparrow 1$: The variance term scales like $\frac{1}{1-\beta}$, so while memory increases, the worst-case bound deteriorates in its constant; careful tuning of α becomes more important.